ANALYTICS REPORT

2026

ITB's customers are loyal and active, but their financial health rises and falls with their own spending pace. Using a year of card data, **we map who is thriving, who is stretched and who is drifting away, and propose non-punitive ways to help each group spend within their means.**

Business Intelligence

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Situation	ITB serves 999 card customers who spent VND 3,244.6 bn across 1.85 M transactions in 2025. Most look healthy on average: 72.5% of customer-months are Stable, and engagement averages 78 out of 100.
Complication	Engagement and health pull apart: the most active, digital customers are the most stretched, while many financially healthy customers barely use the card.
Key Findings	• One driver decides health: spend-to-income explains 81% of the score, far more than job, location or age. • Depth, not reach, is the engagement lever: everyone uses every channel, but only 21% of spend is online, and online share drives engagement most. • Five segments need different support: Healthy Core (299), Emerging Digital (184), Stretched & Engaged (226), Low-Engagement Vulnerable (199) and New / Low-history (91).
Question	How can ITB support customers' financial wellbeing and deepen engagement at the same time, without penalising anyone?
Answer	• Budgeting tools, spend alerts and planning reminders before December for customers who overspend • Cashback and savings goals for healthy but low-use customers . • Digital nudges and financial education: app/QR onboarding for older and new customers, and short money lessons for all with a tough month .

Overview

Busy months mean more purchases, not a higher AOV (Task 1, Q1)

DECEMBER PEAK: Spending hits its yearly high, 2.8x the lowest month

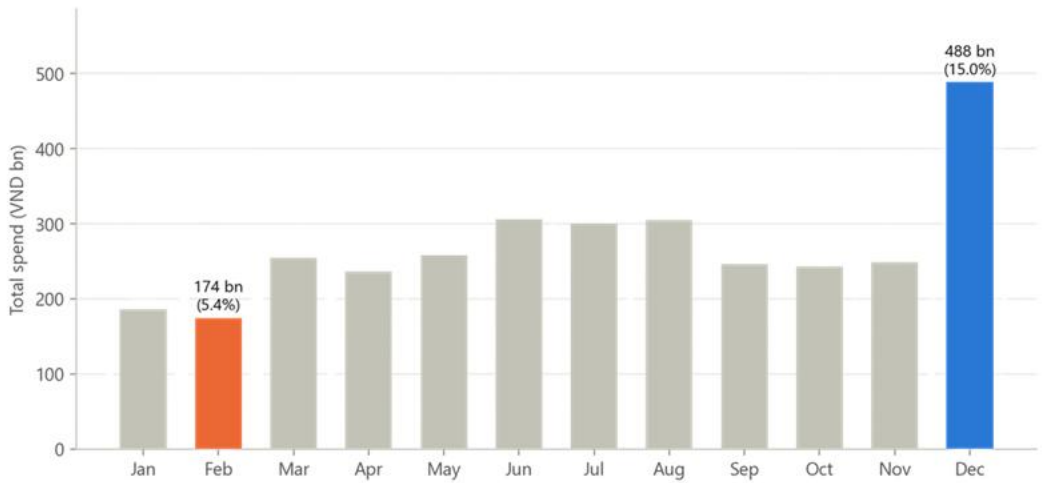


Figure 1. Total spend by month, 2025

- December reaches VND 488 bn (15% of the year) vs 174 bn in February, a gap of VND 314 bn
- Spend-to-income jumps to 1.25 in December, vs 0.45–0.79 in other months: customers spend more than they earn

WHAT DRIVES IT: More transactions, not bigger baskets

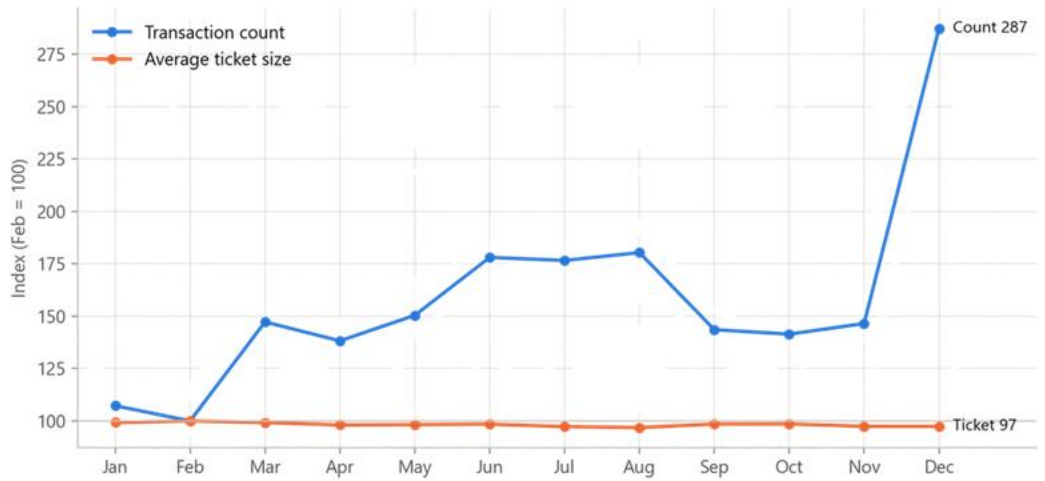


Figure 2. Transaction count and average ticket size, indexed to February = 100

- Transactions rise 2.9x, while the average ticket shrinks 2.6%, so volume explains the whole gap
- Same customers (918 vs 919), each buying 306 vs 106 times; all 14 categories double, and the spending mix stays unchanged

December is the peak spending month because customers buy almost three times as often, not because they buy bigger: each transaction is even 2.6% smaller than in February. These frequent, ordinary purchases add up to overspending in this month.

Overview

Fuel gets swiped most often, but groceries take the biggest share of wallet (Task 1, Q4)

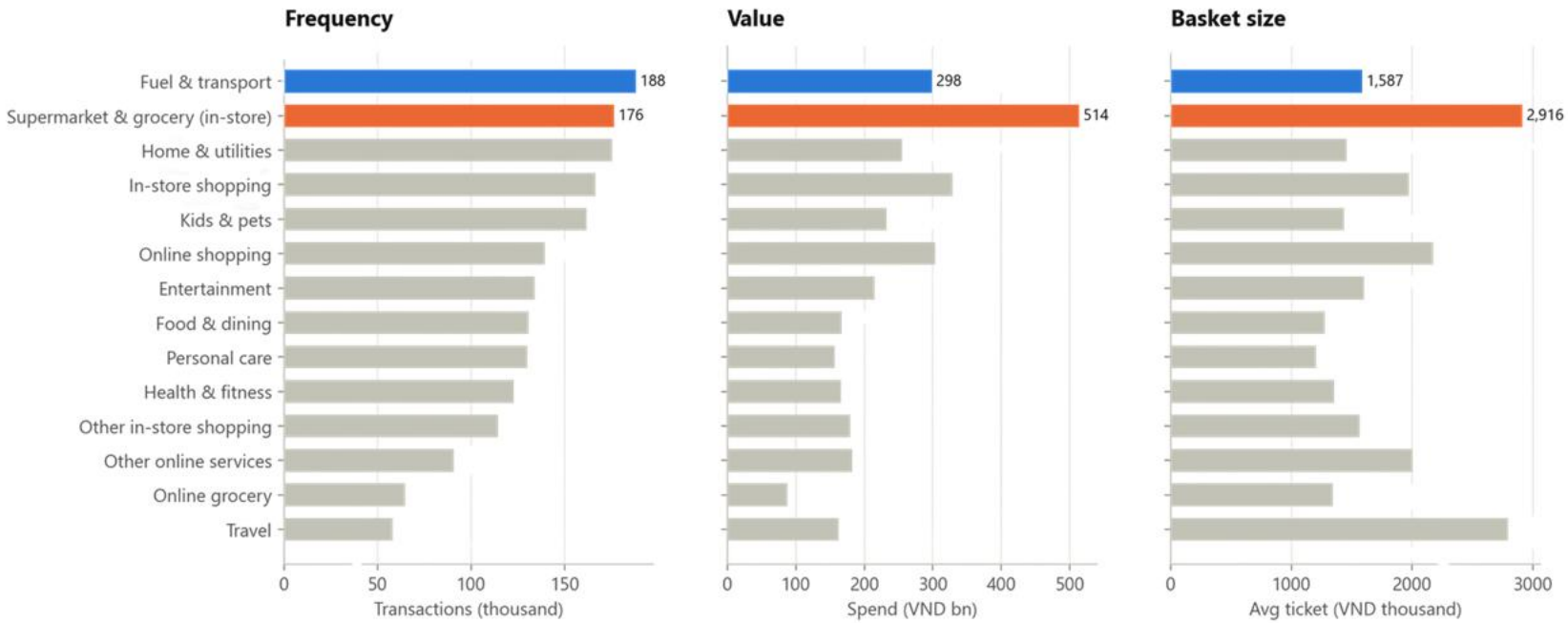


Figure 3. Transactions, spend and average ticket by category

Fuel is a small, repeated top-up; groceries are a basket nearly twice as large. Frequency builds engagement, basket size builds value.

- By transaction count: Fuel & transport, 188,029 transactions (10.2%), average ticket VND 1,586,933.
- By spend: Supermarket & grocery (in-store), VND 513.8 bn (15.8%), average ticket VND 2,916,003, which is 1.84x larger.
- Visit frequency is almost the same (17.1 vs 16.0 transactions per customer-month), and both are POS-heavy (69.8% and 75.1% POS).

Overview

Customers use every channel, and where they live matters less than their age (Task 3, Q2 -Task 1, Q3)

CHANNELS: Everyone tries every channel, but online use stays shallow

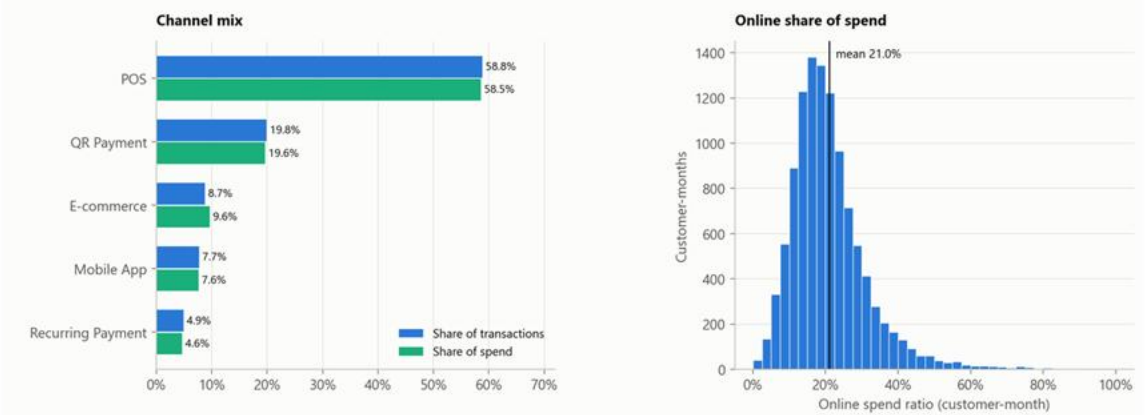


Figure 4. Channel mix and distribution of online spend ratio

- 925 of 999 customers use all five channels, yet POS still carries 59% of transactions
- Only 21% of spend is online (41% including QR); Recurring Payment is just 4.9%, the channel with the most headroom

Adoption is broad but shallow: the issue is depth, not reach. The digital gap is generational, not regional, and no lagging province differs significantly from the rest.

GEOGRAPHY VS AGE: The digital gap is generational, not regional

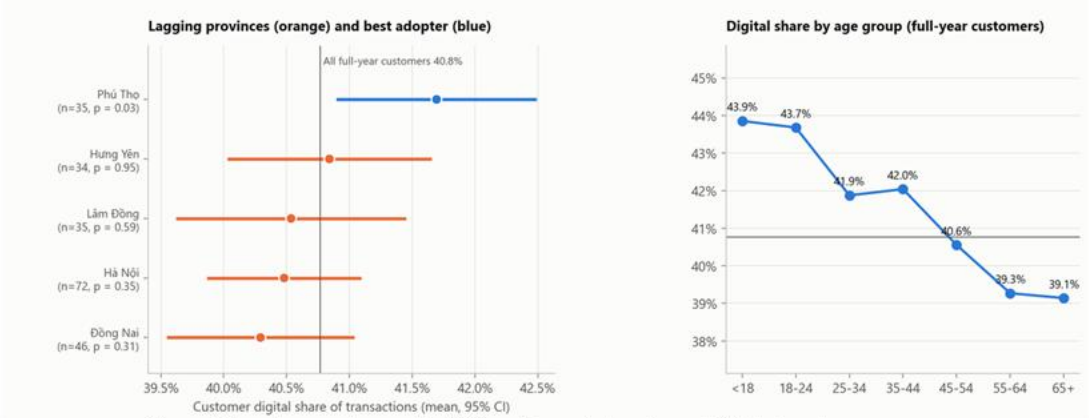


Figure 6. Lagging provinces with 95% confidence intervals and digital share by age group

- Four top-10 provinces trail the national share by only 0.1–0.5 pts, and none is significant ($p \geq 0.31$). Closing the gap would shift only about 1,300 transactions a year
- Digital share falls steadily with age, from 43.9% to 39.1% (rho −0.52, $p < 0.001$)

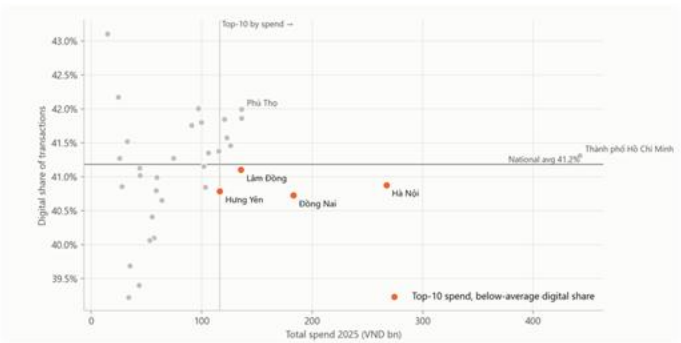


Figure 5. Province spend versus digital share of transactions

Four high - spend provinces sit just below the national line

Financial Health Analysis

Most customers have tough months, and they come with the spending peaks (Task 2, Q1)

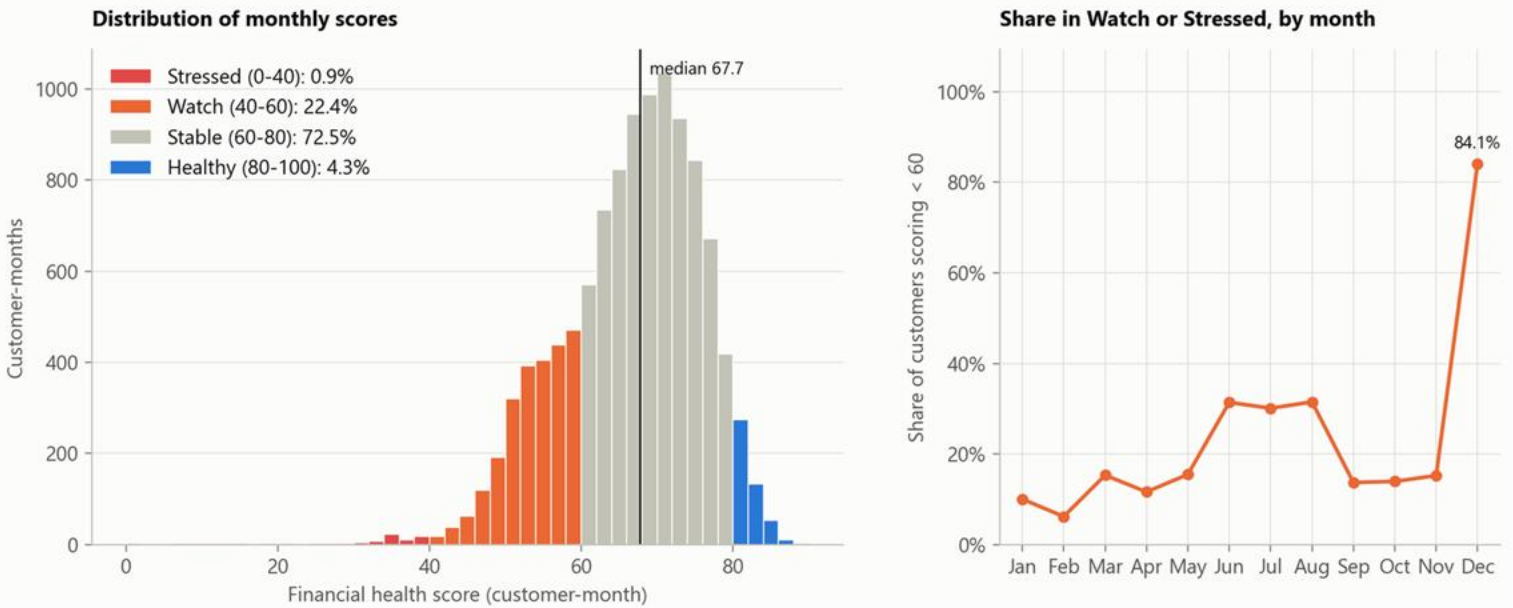


Figure 7. Distribution of monthly health scores and share below 60 by month

Almost every customer goes through a tough month, and financial difficulty is closely tied to spending: in the peak month, December, 84% of customers fall below 60.

- There is a long tail of bad months. 22.4% of months fall to Watch (40–60), which pulls the average down.
- Almost everyone has a tough month. 87% of customers have at least one (according to calculation)
- December is the worst month (84% of customers in a tough month). February is the best

Financial Health Analysis

Tough months are months when customers spend more than they earn (Task 2, Q2)

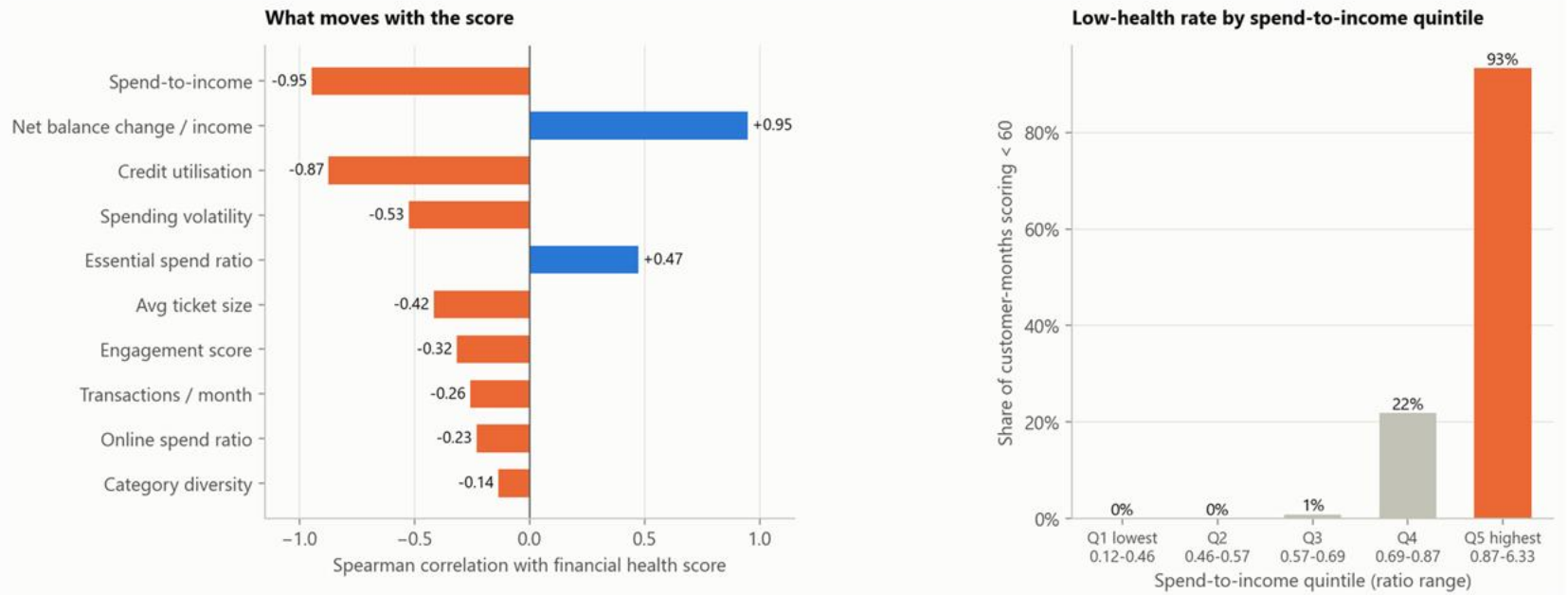


Figure 8. Correlates of the health score and low-health rate by spend-to-income quintile

Low financial health comes mainly from spending too close to income. The more customers spend relative to what they earn, the lower their score tends to be, far more than with any other factor. Irregular spending and a smaller share on essentials add little once this is known.

- One factor dominates. Spend-to-income alone explains 81% of the health score (rho -0.95).
- Credit utilisation (rho -0.87) moves with it (correlation 0.90), so both reflect the same over-spending pressure.
- Risk jumps past a threshold: tough months are almost absent below ~70% of income but reach 93% above 87%.

Financial Health Analysis

In stressed months, discretionary spending takes over the budget and pushes it far past (Task 1, Q2)

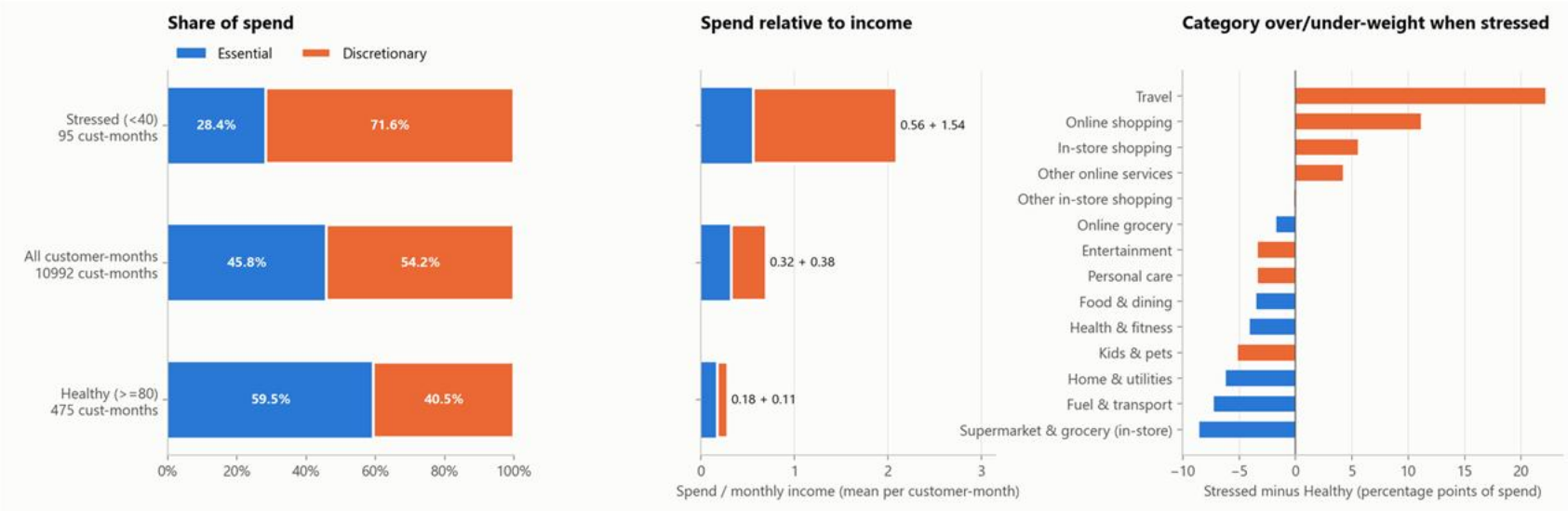


Figure 9. Essential versus discretionary spend by health band: share, level relative to income and category detail

Stressed months are driven by discretionary spending, mainly travel and shopping, not by essentials.

- Discretionary dominates stressed months: 72% of spend vs 41% in healthy months.
- Travel leads the shift (+22.2 pts), then online shopping (+11 pts).
- Discretionary explodes: 1.54x income vs 0.11x, pushing total spend to about 2.1x income.

Financial Health Analysis

It's how people spend, not who they are, and that shapes financial health (Task 2,Q3; Task 1, Q5)

DEMOGRAPHICS: Job and location matter a little, and only through spending

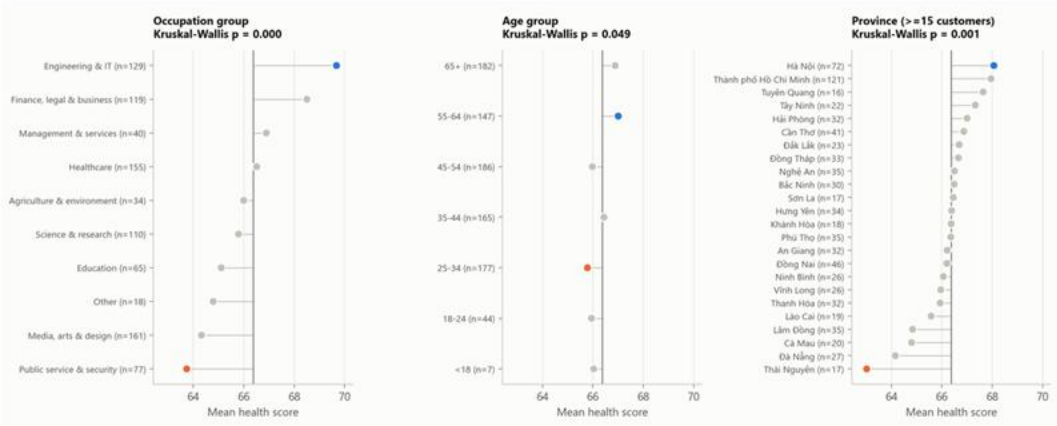


Figure 10. Mean health score by occupation, age group and province (full-year customers)

- Occupation and province shift health by only 5–6 pts (Engineering & IT 69.7 vs Public service 63.7; Hà Nội 68.1 vs Thái Nguyên 63.0), and both gaps track spend-to-income (rho −1.00 and −0.92)
- Spend-to-income alone moves the score by about 25 pts across its quintiles, roughly 4x the largest demographic gap

AGE: Young workers spend the most relative to income, but the gap is small

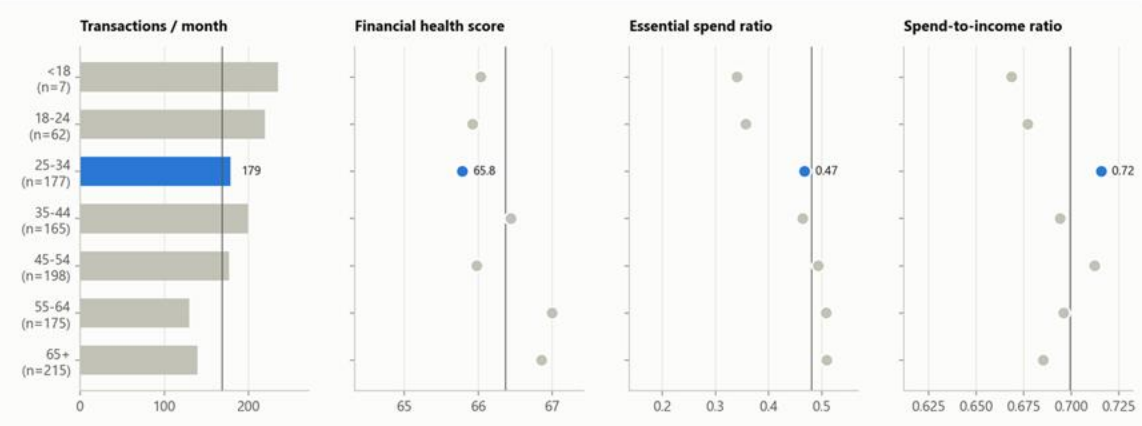


Figure 11. Frequency, health, essential ratio and spend-to-income by age group

- Ages 25–34 transact actively (179/month) yet have the lowest health (65.8), with the highest spend-to-income (0.72) and a lower essential share
- Across all ages, health varies by just 1.2 pts (p = 0.049): age is a weak driver

Occupations and provinces with lower health score lower because their customers spend more relative to income, not because of the job or the location itself. Young working adults spend a lot relative to income and lean less towards essentials, but the age gap is small.

Customer Engagement Analysis

Loyal customers stay engaged, and going digital is what takes them further (Task 3,Q1;Task 3,Q3)

ENGAGEMENT LEVEL: High and even for everyone who stays

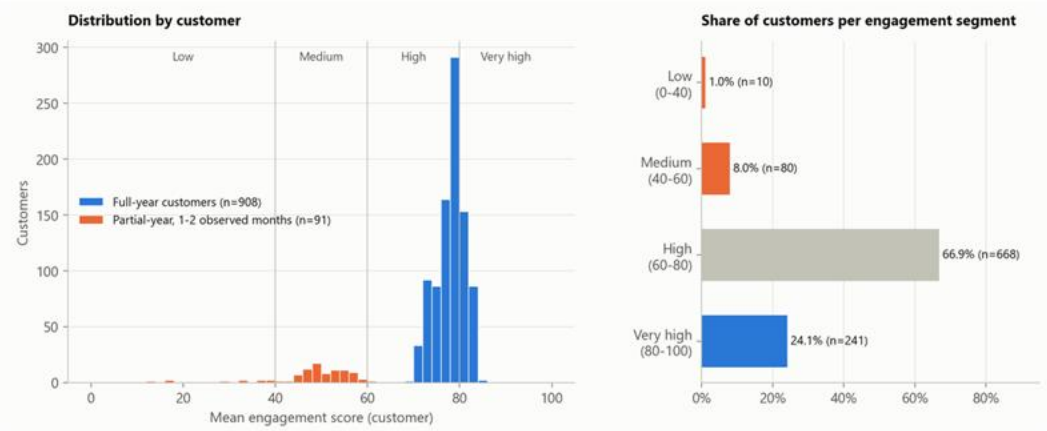


Figure 12. Customer-level engagement distribution and segment shares

- 908 full-year customers average 78.1, all within 70–84; 91% of customers sit in the High or Very high band
- Every Low/Medium scorer has only 1–2 months of data (mean 48.7), a mix of newcomers and leavers rather than proven churn

Engagement among full-year customers is high and even. The engagement challenge is activation of customers with short histories, not retention of established users.

WHAT LIFTS IT: Digital depth drives engagement, and it collapses once customers go quiet

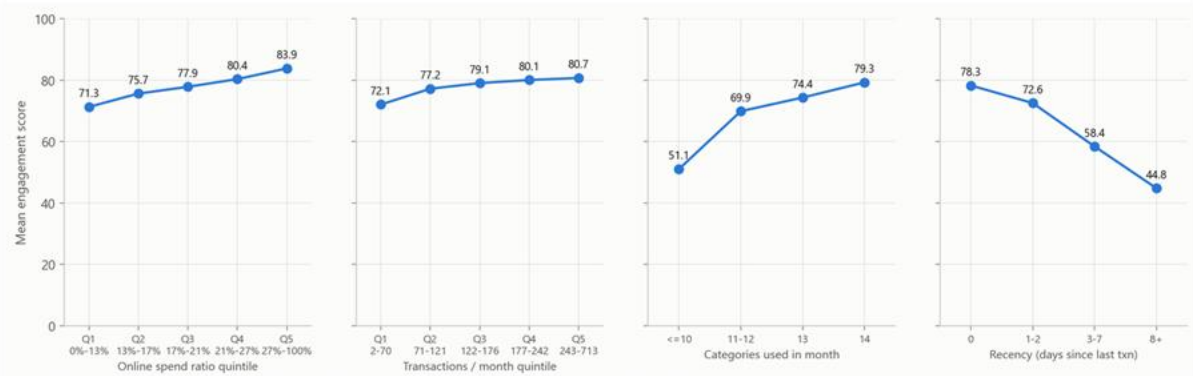


Figure 13. Mean engagement score across bands of four behaviours

- Online share is the strongest lever: engagement rises from 71.3 to 83.9 across quintiles (rho +0.83)
- Categories are already saturated (79% use all 14), while engagement falls to 44.8 after 8+ days without a transaction

Engagement is at heart a measure of digital depth, the only lever that has not saturated. Recency of 8+ days and 10 or fewer categories are two early-warning signals that can be tracked monthly.

The Tension Between Financial Health and Engagement

Most loyal users are the most stretched, while some of healthiest are drifting away (Task 3,Q5; Task 2,Q4)

LOYAL BUT STRETCHED: Daily users who repeatedly run short

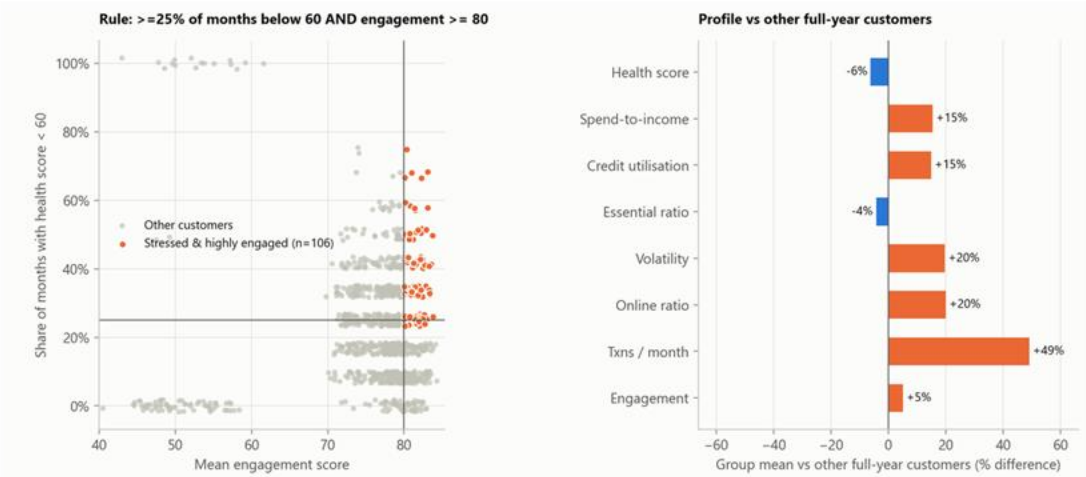


Figure 14. Stressed and highly engaged customers and their profile

- 106 customers (10.6%) drive 16.5% of spend; they make 240 transactions a month (vs 161) at 0.79 spend-to-income (vs 0.69)
- Younger (mean age 39 vs 51); their pressure comes from many small online purchases, not big trips

This group's pressure comes from many small everyday transactions, especially online, unlike the crisis months in section 9 that are driven by large travel expenses.

HEALTHY BUT DRIFTING: Good customers who barely use their card

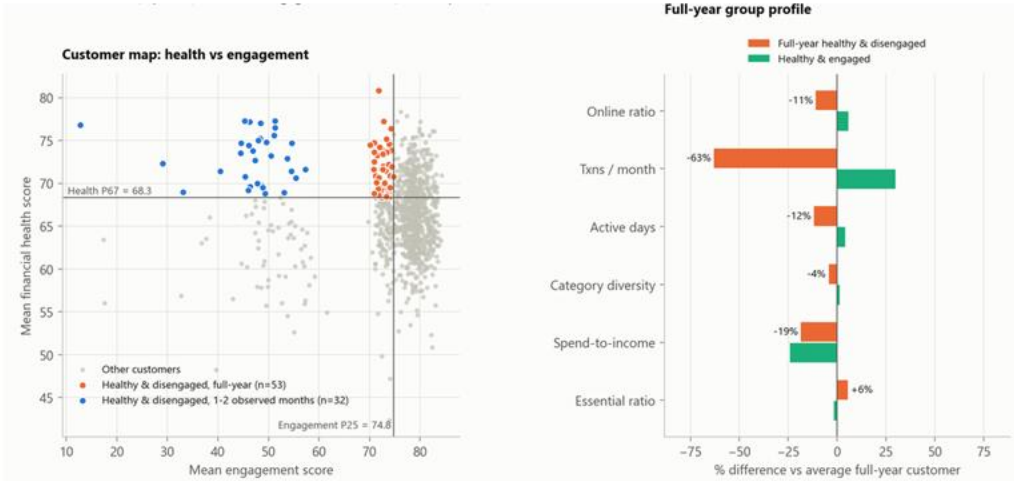


Figure 15. Health vs engagement map and profile of healthy but disengaged customers

- 85 customers (8.5%), of whom 53 are active all year: health 71.5, but 63% fewer transactions and an 11% lower online share
- Older (59% aged 55+): a share-of-wallet problem that matches the age gap in digital use

These are good customers who use their ITB card for only a small part of their spending. This is a share-of-wallet problem, and it matches section 6: digital use falls with age

Data preprocessing

Turning monthly records into clean, comparable behavioural features for segmentation (Task 4, Q1)

Aggregate

- Customer-months → 1 row per customer
- Mean of monthly ratios
- Total spend kept for profiling

Population split

- 908 full-year customers → modelled
- 91 with 1–2 months → rule-based New / Low-history

Quality check

- No imputation needed
- Full transaction file verified
- Near-duplicate feature dropped

Feature engineer

- Financial features
- Engagement features
- Validation features
- Composite indices

Transform

- Winsorize p99
- Log1p
- Z-score scaling
- PCA per block

Model features	
spend_to_income	Monthly spend/income; above 1 means overspending
credit_utilization	Balance / credit limit; reliance on credit
essential_ratio	Share of spend on essentials; spending discipline
spending_volatility	How erratic monthly spending is
txn_count	Transactions per month; usage frequency
active_days	Days with at least one transaction; regularity
online_ratio (log)	Share of spend made online; digital adoption
recency (log)	Days since last transaction; signs of going quiet

Feature selection & PCA

Redundancy check	Pairwise correlation between candidate features	|r| > 0.95 → drop (balance trend vs spend-to-income, r = −0.998)
Block PCA	PCA on standardised features within each block	Eigenvalue > 1 (Kaiser): Financial 2 PCs, Engagement 2 PCs
Block weighting	Standardise PCs, divide by √(number of PCs)	Each block contributes equal variance
Candidate test	Transaction-rhythm block (weekend, night, hour)	Not adopted: hours are fixed by category in synthetic data

- Not used as model inputs: **financial_health_score, engagement_score**

→ Kept as independent benchmarks: our Health Index matches the given score at r = 0.89, and the Engagement Index at r = 0.94

- Demographics: used for description only, never for clustering
- Validation features: share of months below 60, next-month crisis rate

Customer Segmentation

Using K-Means algorithm to group customers into 5 segments (4 K-Means clusters + 1 rule-based group)

Model Chosen

K-means Clustering with Optimal K

Metrics

Silhouette score, Inertia (Elbow)

Input variables

Financial Health:

- Spend_to_Income
- Credit_Utilization
- Essential_Ratio
- Spending_Volatility

Engagement:

- Txn_Count
- Active_Days
- Online_Ratio
- Recency

Standardization

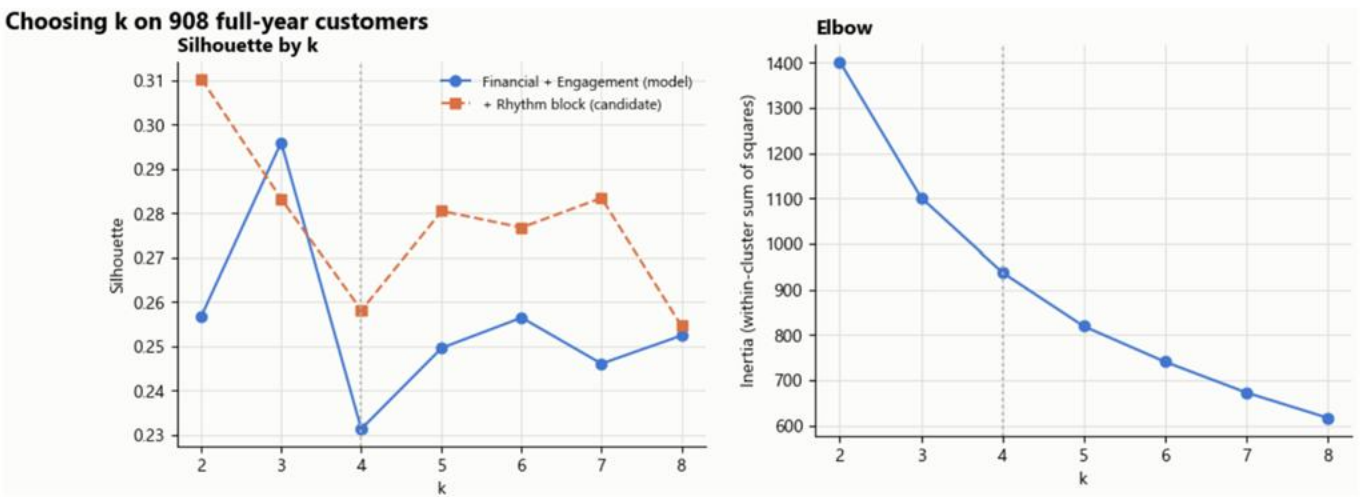
Z-score scaling

999 customers

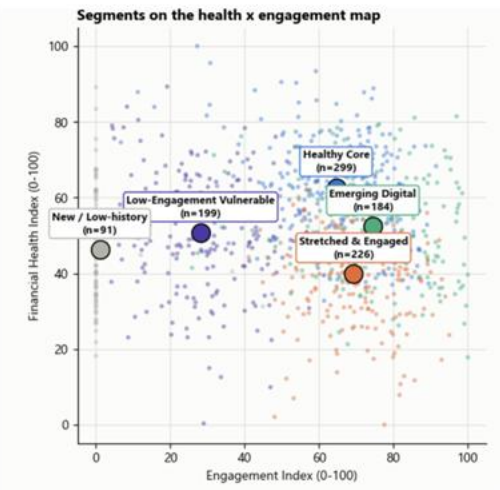
12- month data (908 customers) → Feature Engineering → Clustering

Only 1–2 month data (91 customers) → Rule-based

Determining the number of clusters (K=4)

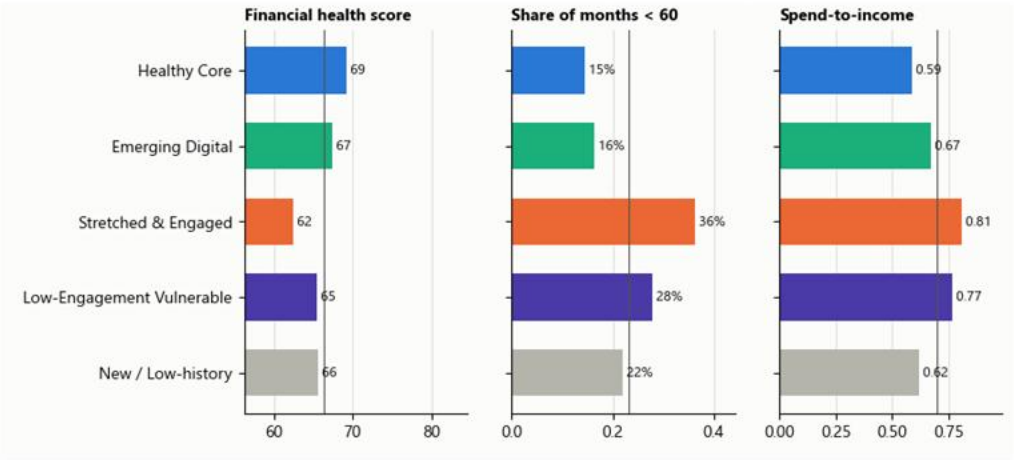


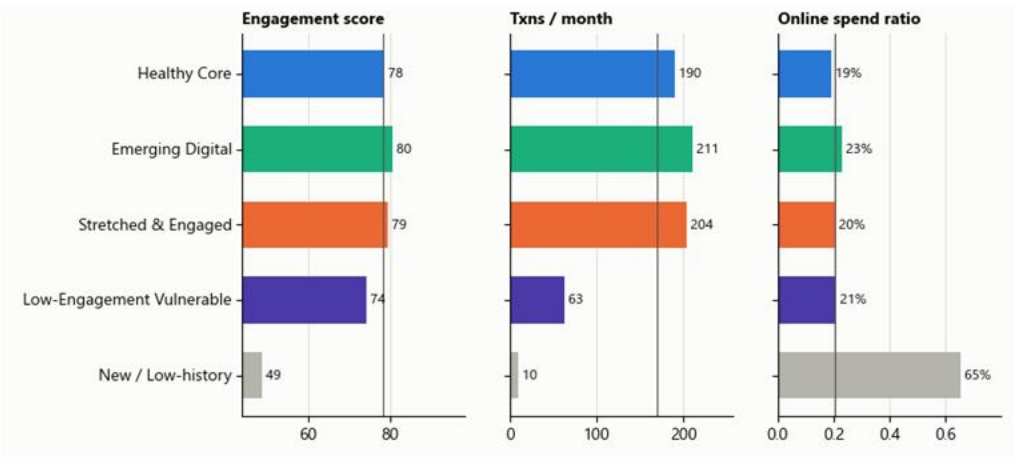
K-means clustering



Customer Segmentation

Identify the characteristics of each cluster and propose tailored strategies accordingly (Task 4, Q2)





	Characteristics of Each Segment
Cluster 0: New / Low-history	Their apparent low activity reflects just 1–2 observed months. They are best treated as customers whose habits are still unknown, rather than labelled inactive or low value.
Cluster 1 : Financially Healthy Core	These customers spend well within their means (59% of income) and account for nearly a third of recorded spending. They appear to be a financially stable customer base with room to save or plan for longer term goals.
Cluster 2 : Emerging Digital Customers	Mostly aged 25–44, they make the most transactions (211/month) and contribute 31.9% of spending. A relatively small group is already generating substantial activity and value through digital channels.
Cluster 3 : Stretched but Highly Engaged	They transact frequently (204/month), but spending consumes 81% of income and their financial health is the weakest. Their activity makes financial strain visible early and creates an opportunity for timely budgeting support.
Cluster 4 : Low Engagement Financially Vulnerable	With only 63 transactions/month and below-average financial health, they show both limited activity and financial pressure. They may be harder to reach through transaction-based messages, so support should be simple and easy to access.

Customer Segmentation

Limitation & Next Steps (Task 4, Q3)

Limitation

#1 **Static averages:** each customer is one 12-month mean vector, so seasonality (e.g. the December spike in Task 1) is averaged away.

#2 **Continuum, not discrete types:** silhouette is modest and HDBSCAN finds no strong density gaps among full-year customers. The four segments are a useful partition of a continuum, not naturally separate populations.

#3 **Partial-year customers are rule-assigned:** 91 customers with only 1-2 months cannot be profiled reliably. Their segment is defined by data length, and they may include both late joiners and early leavers.

#4 **Circularity:** the given financial_health_score is very likely built from the same ratios (Task 2: spend-to-income alone explains 81% of it), so agreement between segments and the score partly reflects construction.

#5 **Synthetic data:** income, limits and balances are synthetic, so only ratios are compared. Hour-of-day features are generation artefacts.

#6 **Fairness:** age differs strongly by segment (a behavioural consequence, not an input). Any campaign should be checked so that older or younger customers are not systematically excluded from beneficial offers.

Future work

#1 Model trajectories (month-to-month migration between segments) instead of yearly averages, and track migration as a KPI.

#2 Try soft assignment (GMM) given the continuum, and report membership probabilities for borderline customers.

#3 Add channel-mix and category-mix features from the transaction file as a further candidate block.

#4 Re-segment quarterly, and A/B-test the recommended interventions per segment.



Recommendation

Six non-punitive interventions

Visual Timelines and Roadmaps

Immediate Actions Timeline

- **Weeks 1–2 (Preparation):**
  - Refresh health & engagement data; set baseline KPIs.
  - Define targeting filters (Health < 60, risk thresholds) & opt-out rules.
- **Weeks 3–4 (Financial Support Launch):**
  - Deploy Budgeting Tools to cap non-essential spend for Health < 60.
  - Launch Real-time Alerts triggered by Spend/Income > 0.80 or Credit Util > 0.70.
  - Roll out Financial Education micro-modules for vulnerable cohorts.

Long-Term Strategies Roadmap

- **Weeks 5–6 (Engagement Support Launch):**
  - Send automated Planning Reminders ahead of due dates.
  - Push Digital Nudges to transition high-spend users to app/QR channels.
  - Offer Suitable Product Suggestions (auto-save, 0% installments).
- **Weeks 7–10 (Monitor & Evaluate):**
  - Track adoption rates, health recovery, and alert fatigue.
  - Decide to scale, refine, or pause interventions based on net user benefit.

Future Analysis Plan

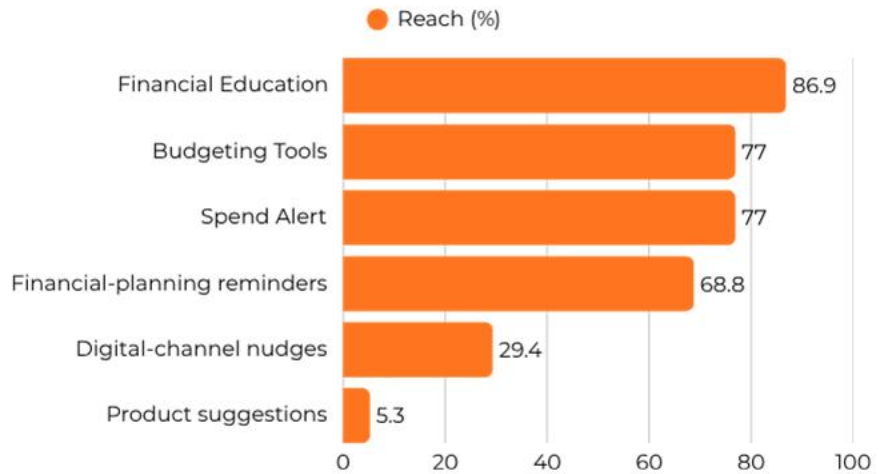
- **Next 6 Months (Predictive Risk & CLV):**
  - Build early-warning models to predict financial distress before health score drops.
  - Measure Customer Lifetime Value (CLV) impact on supported crossover cohorts.
- **Next 12 Months (Advanced Segmentation):**
  - Implement ML-driven segmentation using transaction velocity and merchant mix.
  - Dynamically refine recommendation algorithms based on real-time health trajectories.

Task / Phase	Start Year	Start Qtr	Start Month	Start Day	End Year	End Qtr	End Month	End Day	Tools Covered
Phase 1: Prepare (Weeks 1-2)	2026	Qtr 4	October	1	2026	Qtr 4	October	14	All six tools
Phase 2: Launch Financial Support (Weeks 3-4)	2026	Qtr 4	October	15	2026	Qtr 4	October	28	Budgeting tools; Spend alerts; Financial education
Phase 3: Launch							Novem		Financial-planning reminders;

Recommendation

Six non-punitive interventions

Ranking by reach and by drivers' strength



	Low Impact	High Impact
High Reach	Financial Education Planning Reminders	Budgeting Tools
Low Reach	Digital Nudges Suitable Products	Spend Alerts

Tool Name	Driver Strength (Tác động vào Health Score)	Reach % (% Khách hàng)
1. Budgeting Tools	Very High (Fixes low Essential Ratio & Spend/Income)	86.90%
2. Real-Time Spend Alerts	Very High (Triggers Q5 Spend/Income >0.87 & Credit Util >0.70)	21.90%
3. Financial Education	Medium (Addresses Volatility 1.869 & Cash-Flow)	86.90%
4. Planning Reminders	Low - Medium (Preventive execution, avoids late fees)	68.80%

Recommendation

Six non-punitive interventions

Budgeting tools

Finding

Customers in the Health < 40 band show a severe imbalance in their spending structure: Non-essential spending accounts for a significantly higher proportion of total expenditure compared to essential needs, driving up the Spend-to-Income ratio.

Action

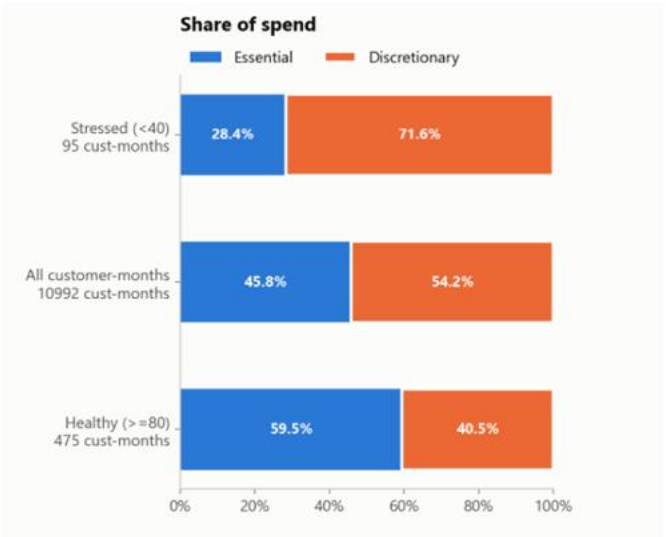
Provide automated Category Budgeting with a focus on Capping Discretionary/Non-essential Spending (e.g., setting a 30% threshold of monthly income for non-essential items using a standard 50/30/20 framework).

Target Group

Customers whose spending exceeded income at least once in the past three months.

Reach

707 customers (77.0% of December customers), prioritizing the 687 who spent more than their income in December.



Financial education

Finding

86.9% of unique customers (869/1,000) experience at least one vulnerable month (Health < 60).

Action

Deliver In-App Micro-Learning Modules (1–2 minute interactive guides & nudges) focusing on:

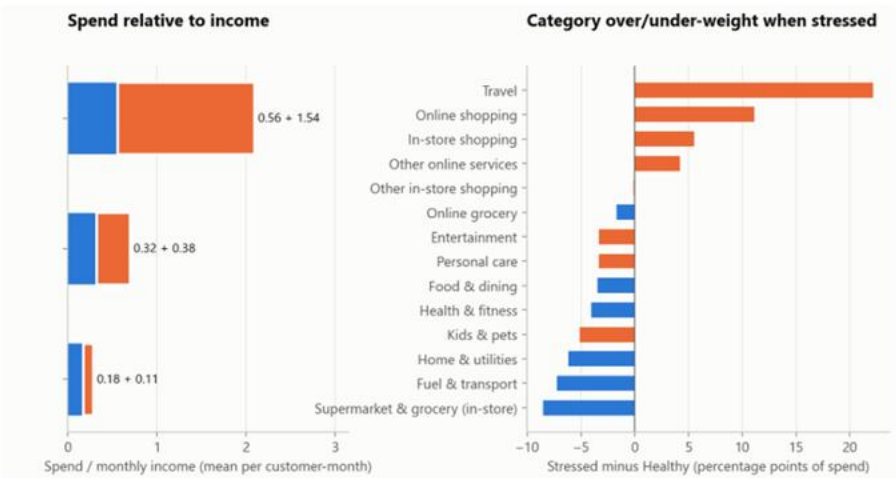
1. Managing discretionary vs. essential cash flows.
2. Building emergency savings buffers to cushion volatility.
3. Smart credit card usage (avoiding high credit utilization penalties).

Target Group

Customers experiencing financial health stress (Financial Health < 60).

Reach

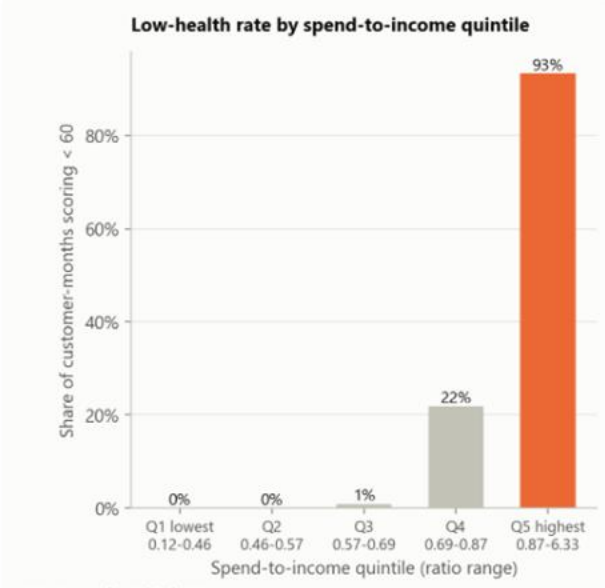
869 customers (86.9%) (All unique customers who recorded at least one month in the vulnerable zone).



Recommendation

Six non-punitive interventions

Spending Alerts



Finding

Customers in the Health < 40 band show a severe imbalance in their spending structure: Non-essential spending accounts for a significantly higher proportion of total expenditure compared to essential needs, driving up the Spend-to-Income ratio.

Reach

707 customers (77.0% of December customers), prioritizing the 687 who spent more than their income in December.

Action

Provide automated Category Budgeting with a focus on Capping Discretionary/Non-essential Spending (e.g., setting a 30% threshold of monthly income for non-essential items using a standard 50/30/20 framework).

Target Group

Customers whose spending exceeded income at least once in the past three months.

Digital-channel nudges

Finding

Among full-year customers, digital transaction share falls from 43.7% for ages 18–24 to 39.1% for ages 65+. Engagement rises from 71.3 to 83.9 across online-spend-ratio quintiles (|ρ| = 0.83).

Action

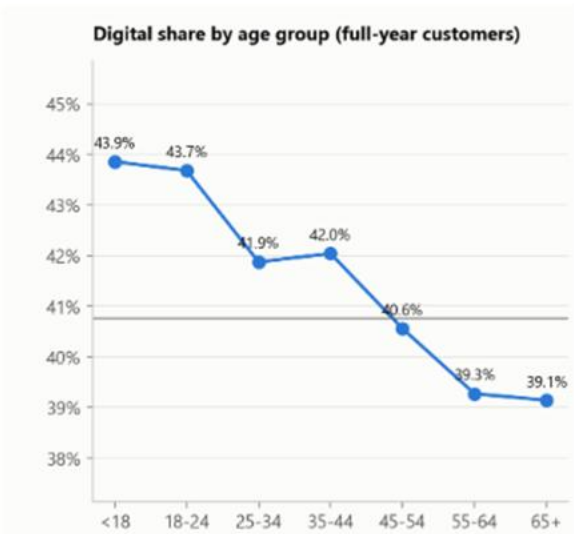
Deepen digital use with assisted QR/app setup and recurring-payment guidance for POS-heavy customers aged 55+. Offer guided onboarding to customers with only 1–2 observed months.

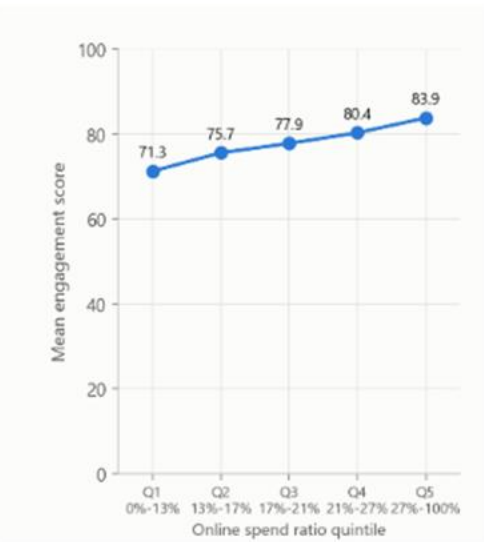
Target Group

Full-year customers aged 55+ with digital transaction share below 40% (203 customers), plus customers in the New/Low-history group (91 customers)

Reach

294 customers (29.4% of 999)





Recommendation

Six non-punitive interventions

Financial-planning reminders

Finding

December spend-to-income reaches 1.25, versus 0.45–0.79 in other months; 84.1% of customers score below 60 in December. December also has 2.9× February's transaction volume, while average ticket size is 2.6% lower ($|\rho| = 0.82$).

Action

Send an October–November reminder to set a year-end budget, then a mid-December check-in showing spend to date and projected month-end spending.

Target Group

Customers whose December spend-to-income ratio is above 1—they spent more than their income that month.

Reach

687 customers (68.8% of 999)

Product suggestions

Finding

Task 3's rule identifies 85 customers with mean health score ≥68.3 and mean engagement score ≤74.8. However, 32 have only 1–2 observed months; among the 53 full-year customers, transactions per month are 63% below the full-year average. This indicates an engagement and share-of-wallet opportunity, not credit risk.

Action

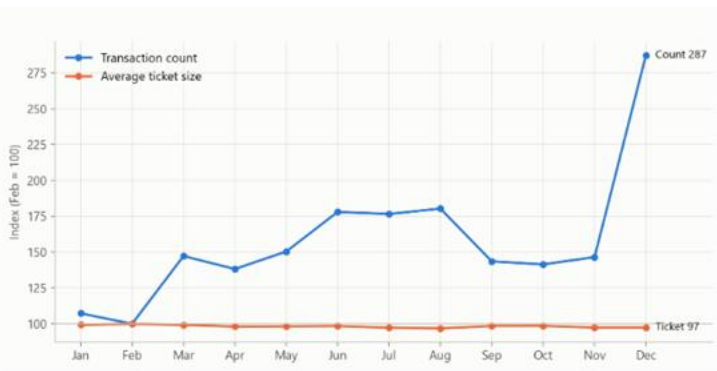
Offer optional savings goals and cashback on everyday categories. Fuel & transport leads transaction frequency (188,029 transactions); supermarket & grocery has the highest spend (VND 513.8bn).

Target Group

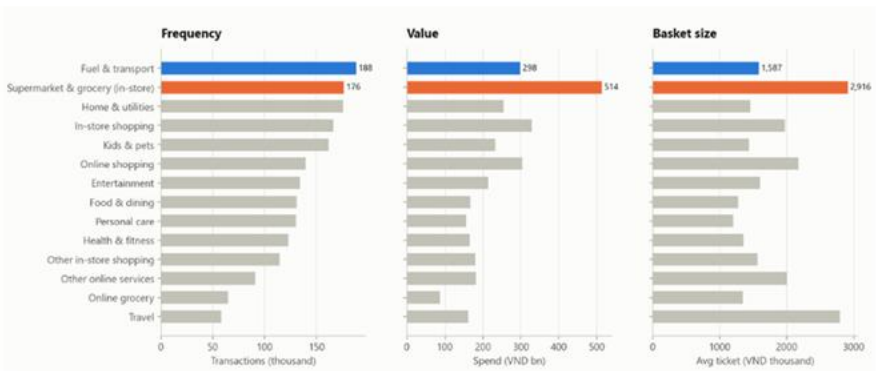
For a sustained low-use product offer, target full-year customers meeting Task 3's rule: mean health score ≥68.3 and mean engagement score ≤74.8.

Reach

53 customers (5.3% of 999)









Thank You